

LLM Sizing of an Equalizer

Ocean's 2

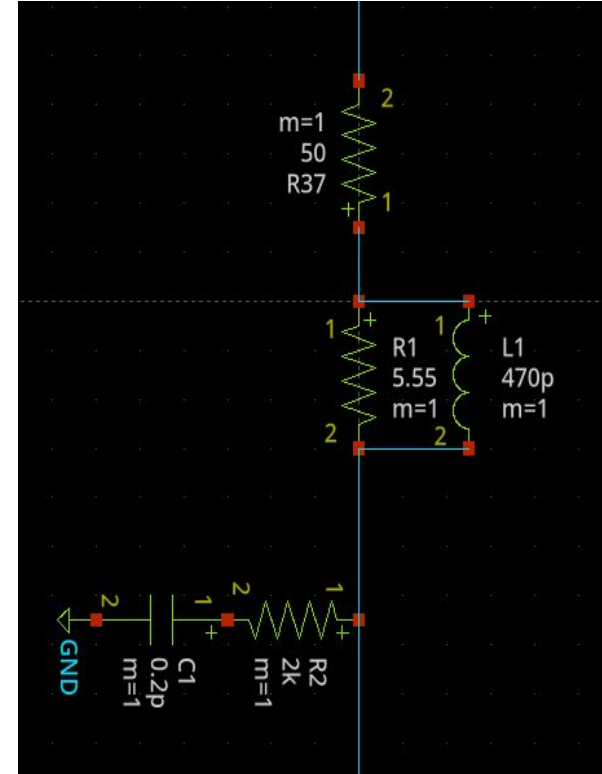
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Literature Review

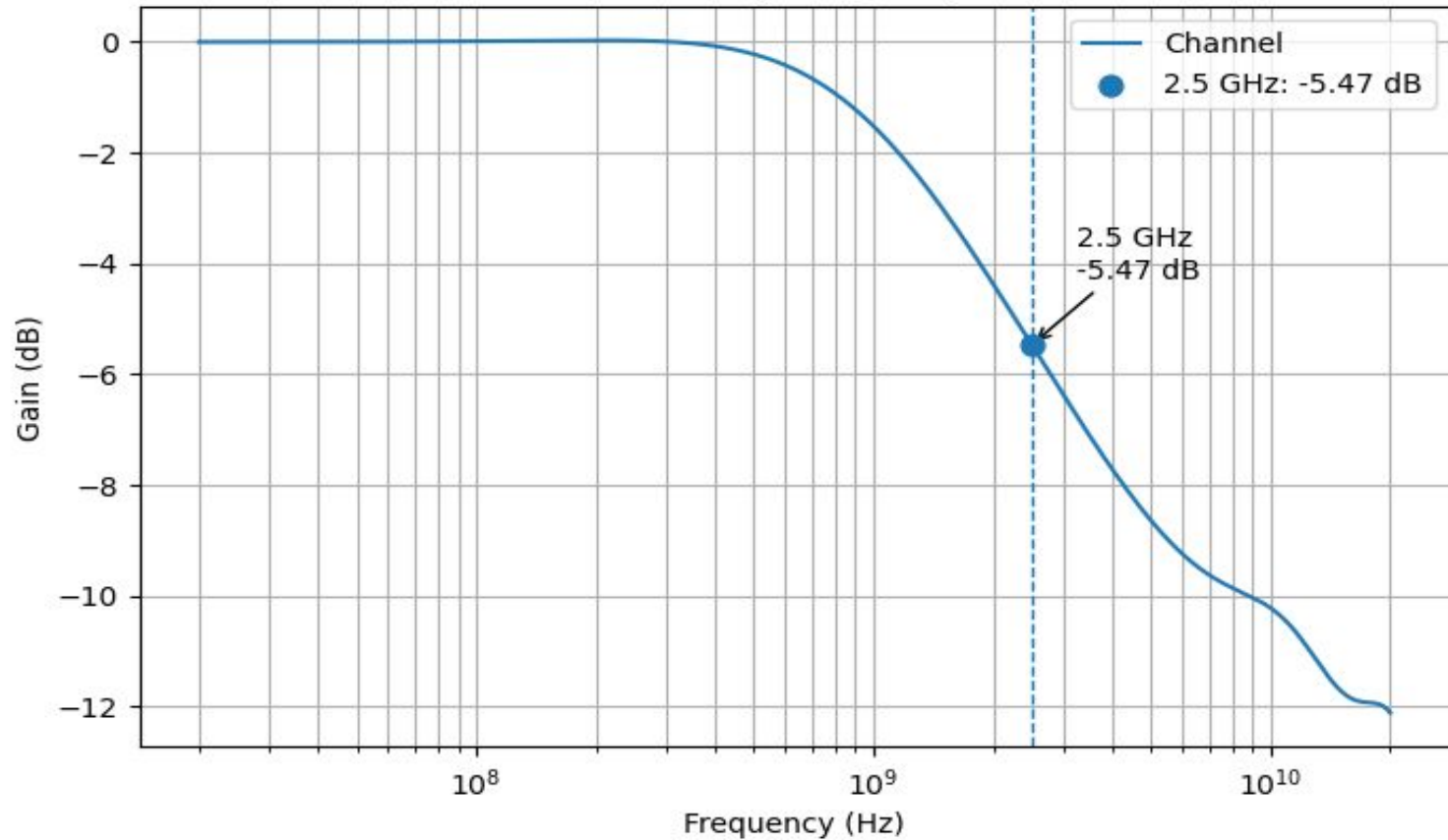
- The Design of an Equalizer (Part one)
- The Design of an Equalizer (Part two)
- EESizer - LLM-based Mixed Signal Circuit Sizing
- CMOS Transistor Sizing

Channel

- Uses 8 cascaded sections of the subsection defined by Razavi, with 100 Ohm I/O resistors.



Channel Frequency Response



CTLE

The equalizer architecture directly following the lossy channel is a Continuous-Time Linear Equalizer (CTLE). The overall signal path follows this sequence:

TX → Lossy Channel → CTLE → DFE Summer → Latch → Output

The following formulas were used to calculate parameters -

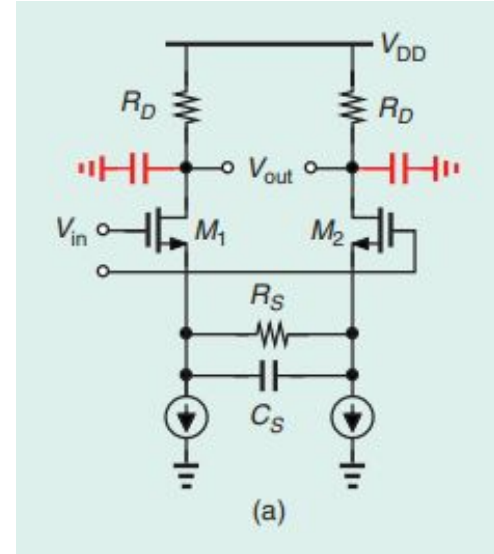
The following formulas were used -

Zero Frequency: $\omega_z = 1/(R_s \cdot C_s)$

Pole Frequency: $\omega_p = (1 + g_m R_s/2)/(R_s \cdot C_s)$

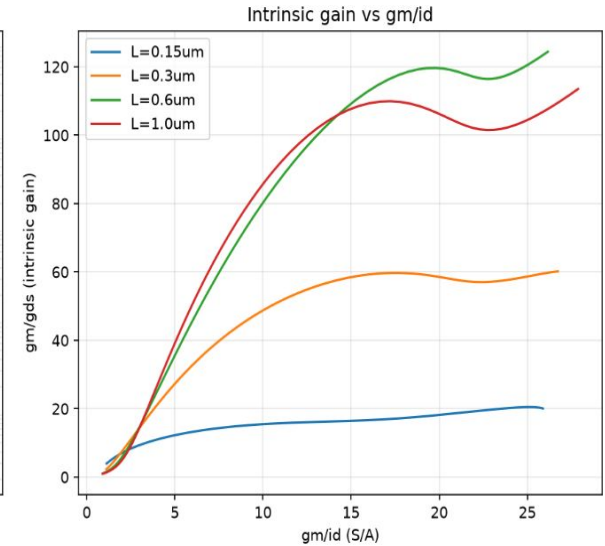
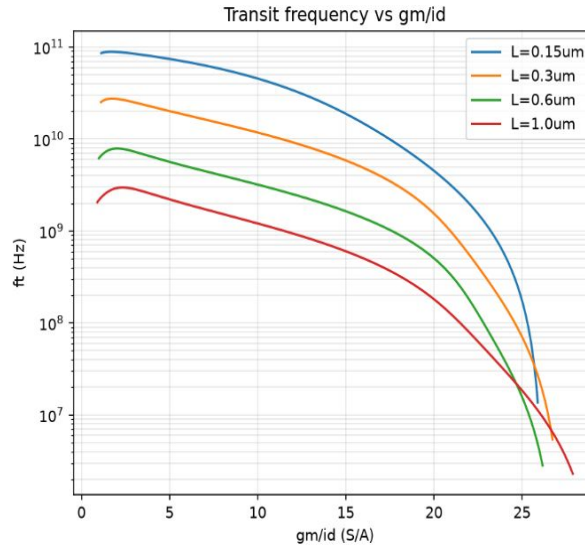
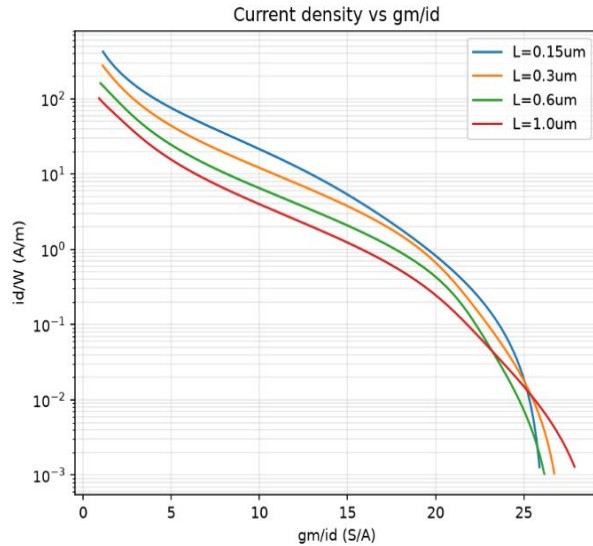
DC Gain: $A_{dc} = (g_m \cdot R_d)(1 + (g_m \cdot R_s/2))$

Peaking Gain: $A = g_m \cdot R_d$



CTLE

Furthermore, the gm/Id methodology was employed to extract the baseline channel width (W) and length (L) for the MOSFETs.



CTLE

gm/id values obtained for initial probe gave us -

$$W = 99.6\mu\text{m}$$

$$L = 0.15\mu\text{m}$$

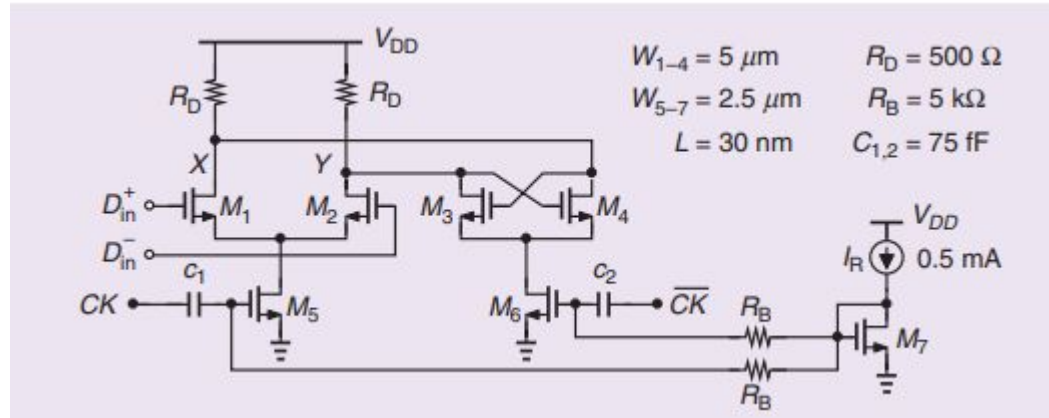
$$R_s = 50.5 \text{ ohms}$$

$$C_s = 3.3\text{e-}12$$

$$R_d = 80 \text{ ohms}$$

$$I_{\text{bias}} = 1.25\text{e-}3$$

DFE

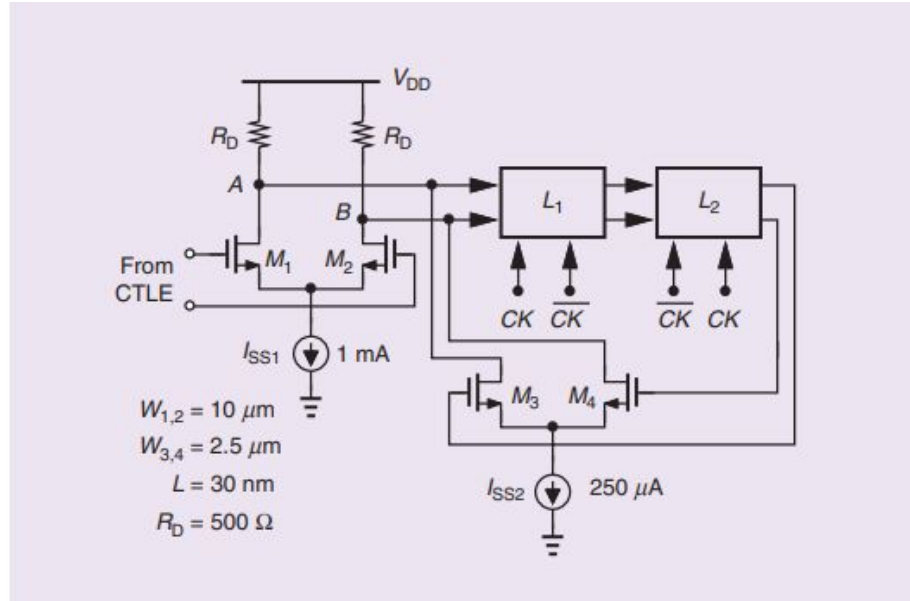


The Current-Mode Logic (CML) latch was designed, taking architectural reference from Behzad Razavi's topology . The parameters were found from the gm/id graphs produced before and shown in the next slide-

DFE

Component	W(um)	L (um)	W/nf
M1-M7, M3-M6	5	0.15	1 um/finger
M4-M5	4	0.15	1 um/finger
R1 and R2	1.41	4.4	3.2Kohm
I_R	-	-	300uA each

DFE

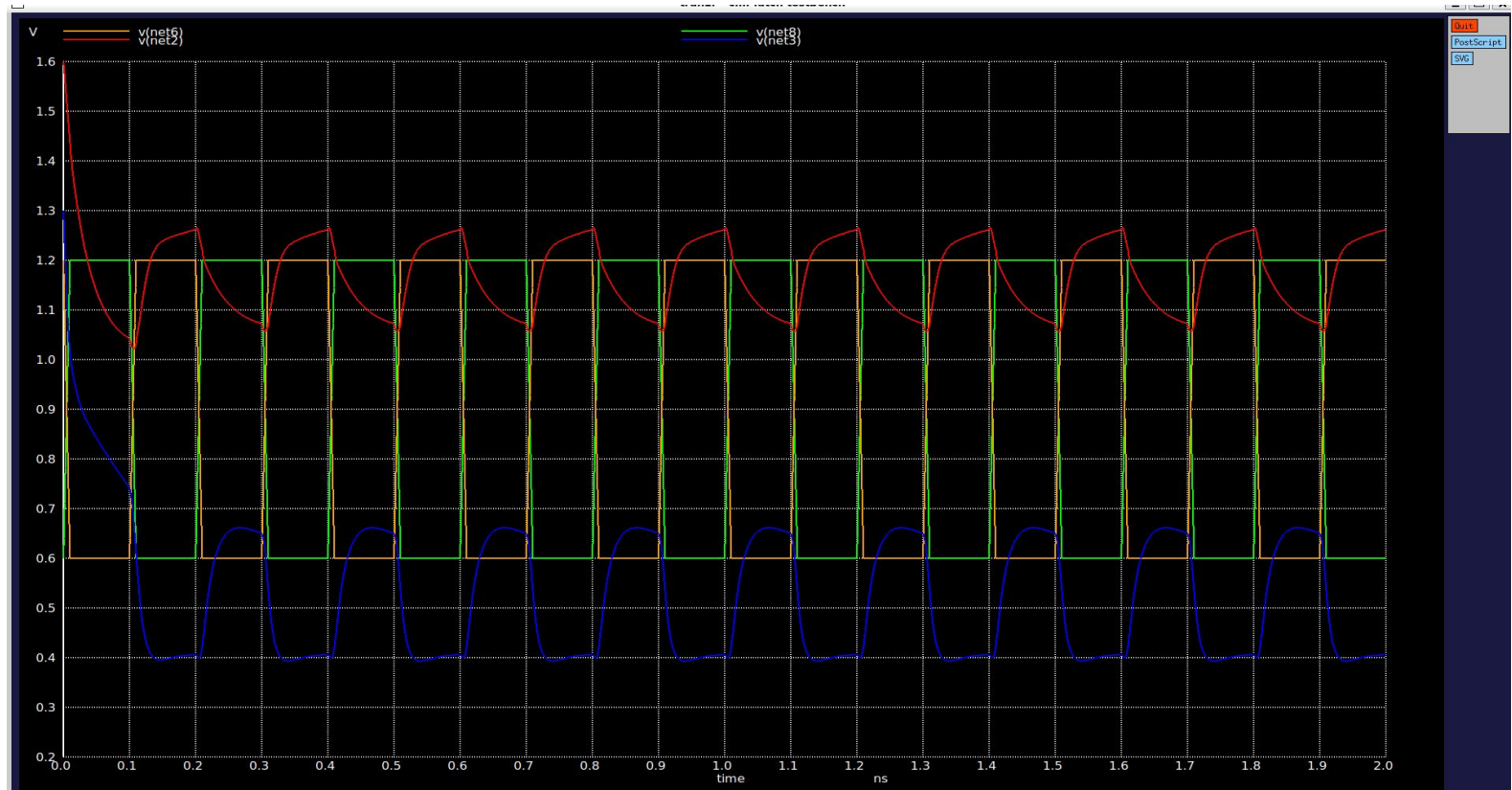


The CML was implemented in the summer. The parameters are as shown in the next slide.

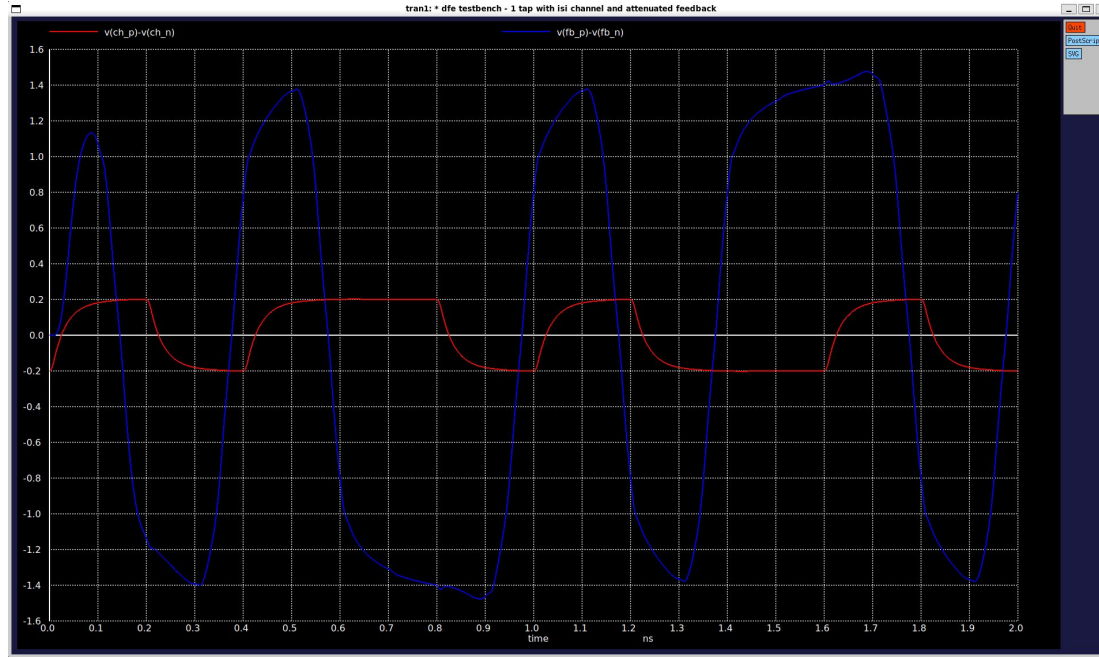
DFE

Component name	W	L	W/nf
M1- M2	5um	150 nm	1um/finger
M3-M4	5 um	150 nm	1 um/finger
Rd	1.41um	4.4um	3.2kilo ohm

DFE



DFE



red is ISI corrupted signal with a small swing of $\pm 0.2V$. The test bench inputs the channel to the dfe. Blue is DFE output with a large swing of $\pm 1.4V$.

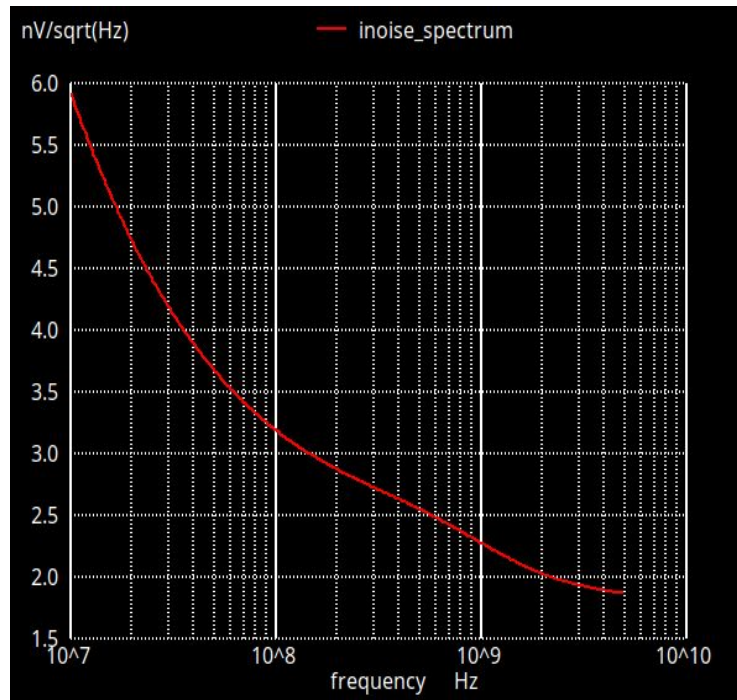
Optimization

The main circuit to optimize is the CTLE, as the DFE is already optimized.

The initial Gm/Id Probes revealed the following :

- HD3 Linearity : -70.08305763580952 dB
- INR = 0.15 mVrms (10MHz to 5GHz)

These are well above spec. So focus optimization on frequency response and eye analysis.



Reward Construction

The reward targets 5 circuit characteristics : Eye height, Eye Width, Boost, Peak Frequency, and DC gain.

Eye Height ≥ 100 mV

Eye Width ≥ 0.4 UI

$3 \leq \text{Boost} \leq 12$ dB

$1.25 \leq \text{pole frequency} \leq 2.5$ GHz

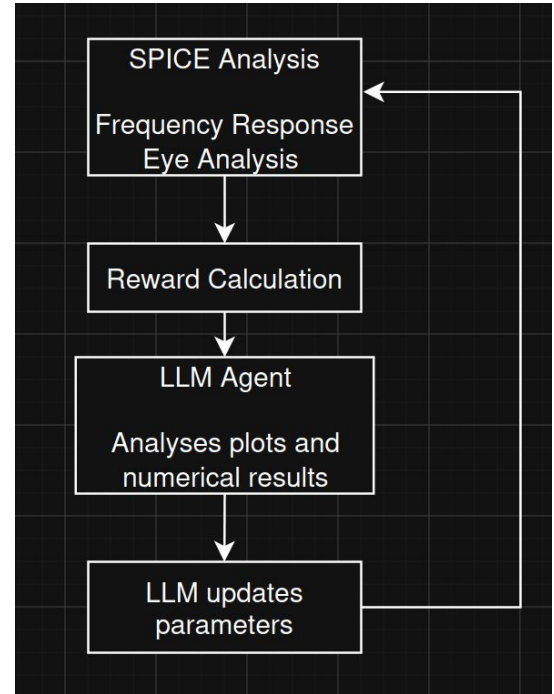
DC gain ≈ 0 dB

LLM Sizing Agent

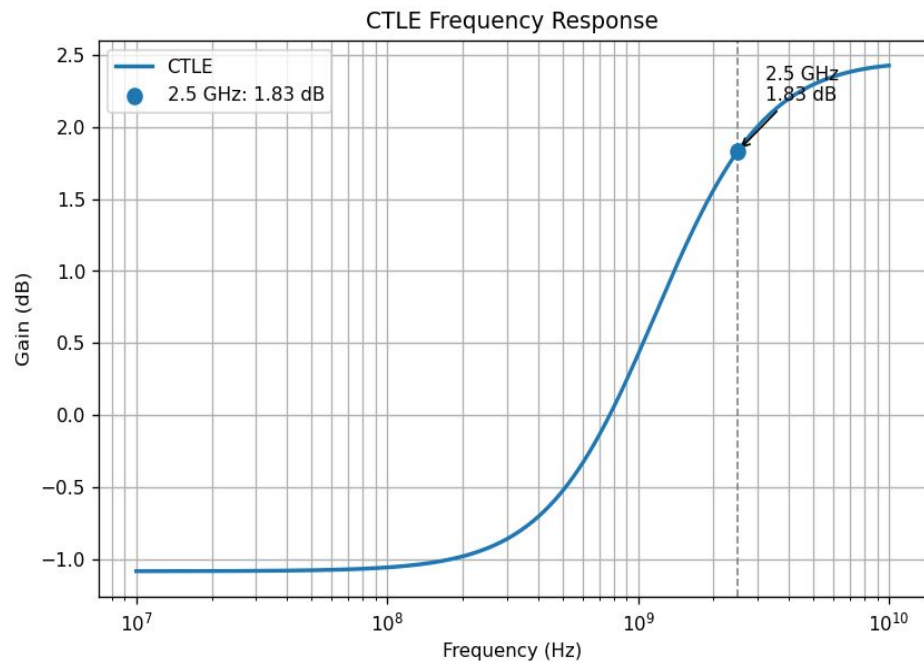
The parameters open to CTLE optimization are W/L, Ibias, Rs, Cs, Rd, and the DFE coefficient.

The initial probe was determined using G_m/I_d , and the initial DFE coefficient was obtained by interpreting the impulse response of the DFE.

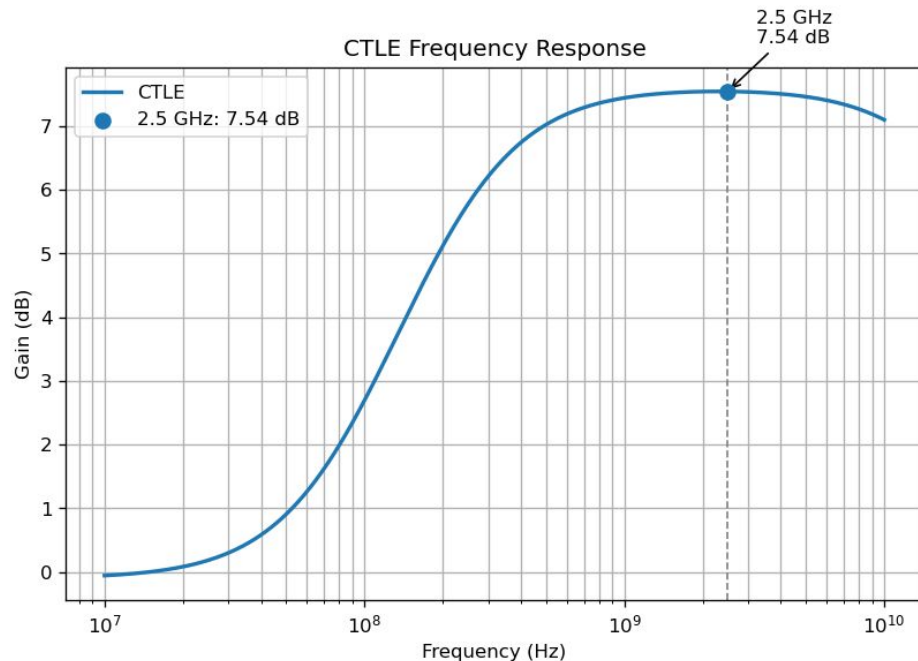
The LLM interprets SPICE results and proposes new parameters.



Results : Frequency Response

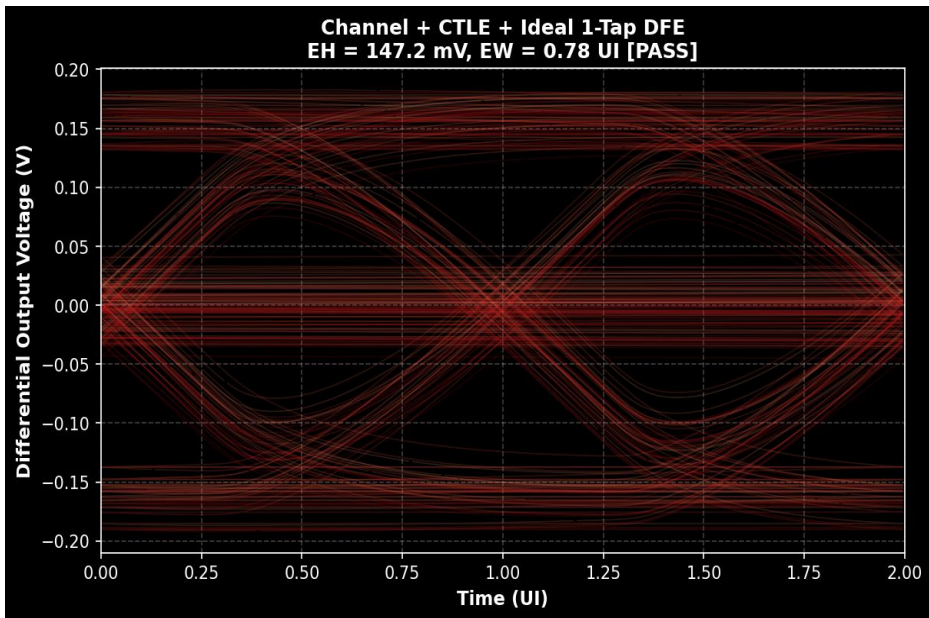


Iteration 0

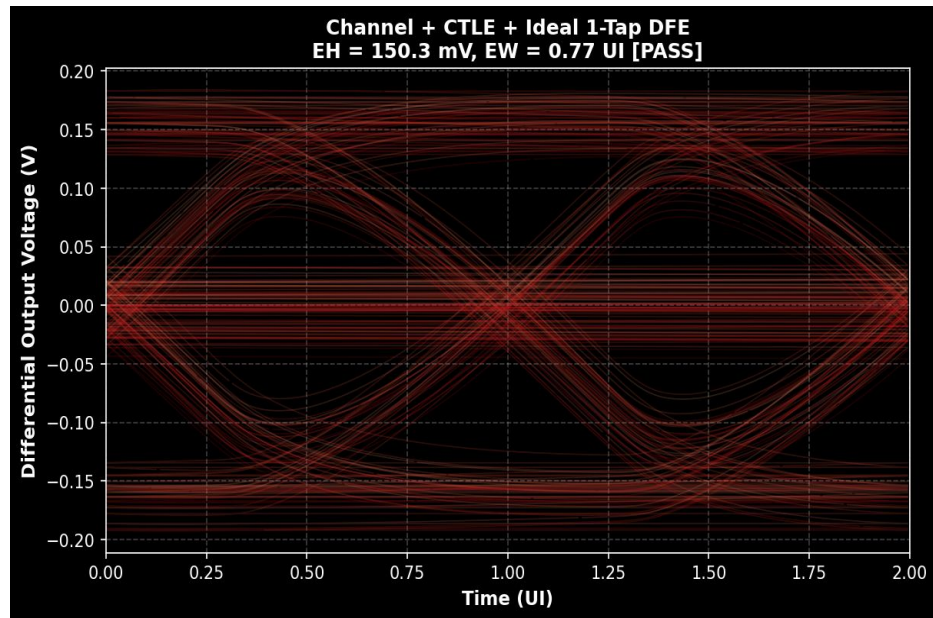


Iteration 10

Results : Eye Opening



Iteration 0



Iteration 10

Results : INR, HD₃

HD₃

- Fundamental: **100 MHz**
- Fundamental amplitude: **34.17 mV**
- 3rd harmonic: **300 MHz**
- 3rd harmonic amplitude: **9.76 μ V**
- Calculated **HD₃ = -70.89 dB**

INR

- Integrated over the specified noise bandwidth
- **Input-referred noise = 0.186 mV RMS**
- Approximately **8× below the 1.5 mV RMS limit**

Characteristic	Final Value	Specification
INR @ 10 MHz	0.186 mV RMS	< 1.5 mV RMS
HD ₃ @ 100 MHz	-70.89 dB	< -30 dB

Cost/Time Analysis

Measured API cost normalized by the number of optimization iterations.

	DeepSeek V4-Flash	Claude Opus 4.5
Cost per Iteration	\$0.01	\$0.0183
Average time per iteration	80.69s	53.50s